

ZIRCON PROJECT SUMMARY

Page 1 ? Vision, Problem, and Value

A modern interlocking engine for a high-stakes industry

ZIRCON is a railway signaling automation platform that converts a structured station description into an interlocking program. In practical terms, it turns the design intent of a station into movement logic and safety requirements that can be consumed by engineering and validation workflows.

The project targets one of the most persistent bottlenecks in rail delivery: the manual, repetitive, and error-prone creation of signaling logic artifacts. Traditional workflows often require engineers to update multiple dependent documents by hand whenever layouts or requirements change. This creates long feedback cycles, hidden inconsistencies, and unnecessary cost.

Why ZIRCON matters

ZIRCON addresses this by making interlocking logic computable from source inputs:

- ? **Layout topography** (‘.zlt’) captures network topology and logical elements.
- ? **Layout geometry** (‘.zlg’) captures distance and spatial positioning.
- ? **Auxiliary data** (‘.zad’) captures context metadata.
- ? **Operational parameters** (‘.zop’) define movement and safety policy.

From these inputs, ZIRCON synthesizes:

- ? route possibilities,
- ? signal behavior,
- ? overlap and protection requirements,
- ? cancellation/clearance delays,
- ? and exportable interlocking outputs (including ‘.xlsx’ artifacts).

This architecture enables faster iterations and safer change management because engineers can regenerate outputs whenever assumptions change.

Core product promise

ZIRCON’s core promise is simple and powerful: **describe once, derive many times**.

By treating station logic as a computable pipeline rather than static documentation, ZIRCON provides:

1. **Speed** ? faster design and validation iterations.
2. **Consistency** ? less divergence between design intent and generated artifacts.
3. **Traceability** ? clear lineage from input encoding to generated output.
4. **Scalability** ? reusable processing flow across multiple station configurations.

Where it is already strong

The repository demonstrates a complete end-to-end path: